The theoretical limit is reached when equality holds.

V.

Reference solution:

Newton's equations for the two-dimensional electrons along x and y are

$$ma_x = -eE - ev_y B - \frac{mv_x}{\tau}\,, \tag{17}$$

$$ma_y = ev_x B - \frac{mv_y}{\tau}\,. \tag{18}$$

In steady state $\boldsymbol{v}$ is time-independent, so $a_x = a_y = 0$. Thus (18) gives

$$v_y = \omega\tau v_x\,, \tag{19}$$

Substitution into (17) gives

$$v_x = -\frac{e\tau E}{m\left(1+\omega^2\tau^2\right)}\,, \tag{20}$$

By definition,

$$j_x = \frac{ne^2\tau E}{m\left(1+\omega^2\tau^2\right)}\,, \tag{21}$$

$$j_y = \frac{ne^2\omega\tau^2 E}{m\left(1+\omega^2\tau^2\right)}\,, \tag{22}$$

Thus a current also flows perpendicular to the electric field.

VI.

Reference solution:

1. The electric field E between the plates is

$$E = E_0 \sin(2\pi z/\lambda_z)\sin(2\pi x/\lambda_x - \omega t)\,. \tag{1}$$

The plates are ideal conductors, so at z=0 and d, E=0. Hence

$$2\pi d/\lambda_z = m\pi, \quad \lambda_z = 2d/m, \quad m = 1,2,3\ldots\,. \tag{2}$$

2. Using the relation below, the wave in (1) can be

$$\sin\alpha\sin\beta = \frac{1}{2}\left[\cos(\alpha-\beta) - \cos(\alpha+\beta)\right], \tag{3}$$

expressed as the superposition of two plane electromagnetic waves: